

CLINICAL TRIAL

Tenecteplase Thrombolysis for Stroke up to 24 Hours After Onset With Perfusion Imaging Selection: The CHABLIS-T II Randomized Clinical Trial

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BACKGROUND: Whether it is effective and safe to extend the time window of intravenous thrombolysis up to 24 hours after the last known well is unknown. We aimed to determine the efficacy and safety of tenecteplase in Chinese patients with acute ischemic stroke due to large/medium vessel occlusion within an extended time window.

METHODS: Patients with ischemic stroke presenting 4.5 to 24 hours from the last known well, with a favorable penumbral profile and an associated large/medium vessel occlusion, were randomized 1:1 to either 0.25 mg/kg tenecteplase or the best medical treatment. A favorable penumbral profile was defined as a hypoperfusion lesion volume to infarct core volume ratio >1.2, with an absolute volume difference >10 mL, and an ischemic core volume <70 mL. The primary outcome was the achievement of major reperfusion without symptomatic intracranial hemorrhage within 24 to 48 hours post-randomization. Major reperfusion was defined as the restoration of blood flow of >50% of the involved ischemic territory. Secondary outcomes included recanalization, infarct growth, major neurological improvements, change in the National Institutes of Health Stroke Scale score, hemorrhagic transformation within 24 to 48 hours, systemic bleeding at discharge, and modified Rankin Scale (score 0–1, score 0–2, score 5–6, and modified Rankin Scale distribution) at 90 days. The comparison of the primary outcome between groups was conducted using modified Poisson regression with a log-link function and robust error variance, adjusted for time from the last known well to randomization, the site of vessel occlusion, and planned endovascular treatment.

RESULTS: Among 224 enrolled patients, 111 were assigned to receive tenecteplase and 113 to receive the best medical treatment (including 23% [n=26] of participants who received intravenous tissue-type plasminogen activator). The mean (SD) age of the tenecteplase group and the best medical treatment group was 64.2 (10.4) and 63.6 (11.0) years old, with 72.1% (n=80) and 70.8% (n=80) male enrolled, respectively. A proportion of 54.9% (n=123) of patients were transferred to the catheter room for preplanned endovascular treatment. The primary outcome occurred in 33.3% (n=37) of the tenecteplase group versus 10.8% (n=12) in the best medical treatment group (adjusted relative risk, 3.0 [95% CI, 1.6–5.7]; $P=0.001$). Tenecteplase significantly increased the recanalization rate compared with the best medical treatment (35.8% [n=39] versus 14.3% [n=16], adjusted relative risk, 2.5 [95% CI, 1.4–4.4]; $P=0.002$). There were no significant differences in clinical efficacy outcomes or rates of hemorrhagic transformation between the groups.

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CONCLUSIONS: Administered at a dose of 0.25 mg/kg intravenously, tenecteplase increased reperfusion without symptomatic intracranial hemorrhage in patients with ischemic stroke selected by imaging in late-time window treatment but did not change clinical outcomes at 90 days.

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GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: intracranial hemorrhages ■ ischemic stroke ■ reperfusion ■ tenecteplase ■ thrombolytic therapy

Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
CHABLIS-T	Chinese Acute Tissue-Based Imaging Selection for Lysis in Stroke–Tenecteplase
CTA	computed tomography angiography
CTP	computed tomography perfusion
DT	delay time
EVT	endovascular treatment
IQR	interquartile range
LAA	large artery atherosclerosis
mRS	modified Rankin Scale
NIHSS	National Institutes of Health Stroke Scale
siCH	symptomatic intracranial hemorrhage
TIMELESS	Tenecteplase in Stroke Patients Between 4.5 and 24 Hours
TRACE-III	Tenecteplase Reperfusion Therapy in Acute Ischemic Cerebrovascular Events–III

hours of symptom onset.^{6–10} Yet data on the efficacy of tenecteplase in the late window are limited. The recently published TRACE-III trial (Tenecteplase Reperfusion Therapy in Acute Ischemic Cerebrovascular Events–III) has managed to demonstrate the benefit of tenecteplase treatment in Chinese patients without preplanned EVT.¹¹ As for patients accessible to EVT treatment, the TIMELESS trial (Tenecteplase in Stroke Patients Between 4.5 and 24 Hours) did not show a significant benefit of tenecteplase treatment, with reperfusion rates not significantly differing between tenecteplase and placebo groups post-EVT.¹² Our prior phase IIa trial indicated that tenecteplase at a dose of 0.25 mg/kg holds promise for achieving significant reperfusion without symptomatic intracranial hemorrhage (siCH) in late-window patients, selected based on vessel occlusion and perfusion imaging showing salvageable tissue.¹³ Based on the previous findings of the CHABLIS-T trial (Chinese Acute Tissue-Based Imaging Selection for Lysis in Stroke–Tenecteplase), the CHABLIS-T II trial aimed to assess whether tenecteplase could improve reperfusion without siCH in late-window patients presenting with large or medium vessel occlusion and a favorable penumbral profile as compared with the best medical treatment, in a patient population with optional EVT.

Reperfusion therapy, including intravenous thrombolysis and endovascular treatment (EVT), is the most effective treatment for patients with acute ischemic stroke (AIS). Extending the time window for reperfusion therapy and thus making more patients with AIS eligible for reperfusion therapy has been one of the research priorities in stroke for decades. Through advanced imaging selection, the time window for EVT has been extended to 24 hours after the last known well.^{1–4} However, compared with EVT trials using advanced imaging in the early window, outcomes in the late window for patients with AIS seem less favorable. This discrepancy could be due to the lower percentage (<10%) of late-window patients with AIS receiving alteplase treatment compared with nearly 100% in the early window.^{3–5} This disparity underscores the potential for intravenous thrombolysis to improve outcomes for late-window patients.

Tenecteplase has been shown to be noninferior to alteplase in treating patients with AIS within 4.5

METHODS

Data Availability

Data from the CHABLIS-T II study will be made available at the request of qualified investigators whose proposal of data use has been approved by an independent review committee.

Study Design

The CHABLIS-T II was a multicenter, prospective, block-randomized, open-label, blinded-end point, phase IIb study conducted at 23 stroke centers across mainland China. The detailed trial protocol has been published¹⁴ and is accessible in the Supplemental Material. This study was approved by the ethics committees of the leading site, Huashan Hospital, Fudan University, and all other participating sites. This study was supervised by an independent data safety monitoring board. The study followed the Consolidated Standards of Reporting Trials reporting guideline.

Participants

Eligible patients ranged from 18 to 80, with a premorbid modified Rankin Scale (mRS) of 0 to 2. They presented with AIS within 4.5 to 24 hours after last known well and showed a clinically significant acute neurological deficit, as judged by the investigators and quantified by a National Institutes of Health Stroke Scale (NIHSS) score. Eligibility was also required to meet the dual-target criteria on baseline multimodal imaging. The dual-target criteria were defined as large or medium vessel occlusion in the anterior circulation on baseline computed tomography angiography (CTA), and a favorable penumbral mismatch profile on baseline computed tomography perfusion (CTP). Large or medium vessel occlusion encompassed occlusions in the extracranial or intracranial segments of the internal carotid artery, the first or second segments of the middle cerebral artery, and the first or second segments of the anterior cerebral artery on CTA. In cases of tandem occlusions, the distal occlusion site was considered the culprit for assessing the efficacy of tenecteplase. A favorable penumbral mismatch profile on CTP was determined by a hypoperfusion lesion volume (delay time [DT] >3 seconds) to infarct core volume (relative cerebral blood flow <30%)^{15,16} ratio >1.2, with an absolute volume difference >10 mL, and an ischemic core volume under 70 mL, which was generated by an automated real-time perfusion volumetric software, AutoMISar (Apollo Medical Imaging Technology, Melbourne, Victoria, Australia).

Comprehensive inclusion and exclusion criteria are detailed in the Supplemental Material. Written informed consent was obtained from all eligible patients or their legal proxies before randomization.

Randomization and Masking

Eligible patients, after meeting all inclusion and exclusion criteria, were randomized in a 1:1 ratio to either receive intravenous tenecteplase (CSPC Recomen Pharmaceutical Co, Ltd) or the best medical treatment. This randomization process was conducted via a centralized website, Easy Random Trial, developed by Trial Data Pharmaceutical Technology (Shanghai) Co, Ltd, using permuted blocks. Stratification was based on the interval from the last known well to randomization (4.5–9 hours versus 9–24 hours). Treatment allocation was open-label, with subsequent radiological and clinical evaluations performed by raters blinded to the allocation. The data safety monitoring board had access to unblinded grouped data for monitoring purposes.

Procedures

Patients in the tenecteplase group were administered a 0.25 mg/kg dose (up to a maximum of 25 mg) as a bolus over 5 to 10 seconds, followed by a 2 mL saline flush. The control group received the best medical treatment, excluding tenecteplase thrombolysis. Alteplase treatment within the time window of the Extending the Time for Thrombolysis in Emergency Neurological Deficits study was permitted.¹⁷ Emergent EVT was optional for all randomized patients based on the judgment of local neurointerventionalists. EVT could be averted when the initial catheter angiography showed a modified Treatment In Cerebral Ischemia score of 2b to 3. All patients were provided guideline-based intensive care for AIS, including intravenous thrombolysis and EVT.¹

Outcomes

The primary outcome combined efficacy and safety, defined as major reperfusion at initial catheter angiography or on a 4- to 6-hour repeat CTP, without sICH occurred 24 to 48 hours after randomization. Major reperfusion was defined as the restoration of blood flow of >50% of the involved ischemic territory. For patients without preplanned emergent EVT, major reperfusion was defined as the hypoperfusion lesion volume (DT >3 seconds) of the 4- to 6-hour repeat CTP decreased to <50% of the hypoperfusion lesion volume of the baseline CTP. For patients with preplanned emergent EVT, major reperfusion was assessed as a modified Treatment In Cerebral Ischemia score 2b to 3 at the initial catheter angiography.¹⁸ sICH was defined as any intracranial hemorrhage causing a ≥ 4 -point increase in the NIHSS score from baseline.

The secondary radiological efficacy outcomes consisted of recanalization at the initial catheter angiography or 4- to 6-hour CTA reconstructed from CTP and infarct growth at 3 to 5 days. The secondary clinical efficacy outcomes were excellent functional outcome (mRS score 0–1), good functional outcome (mRS score 0–2), and mRS distribution at 90 days; major neurological improvement (defined as a ≥ 8 -point reduction in the NIHSS or an NIHSS score of 0–1) and change in the NIHSS score at 24 to 48 hours. The secondary radiological safety outcomes were parenchymal hematoma type 2, sICH, and any intracranial hemorrhage at 24 to 48 hours post-randomization. The secondary clinical safety outcomes were poor functional outcome (mRS score 5–6) at 90 days and systemic bleeding before discharge.

All of the source imaging underwent centralized analysis in a core laboratory. Baseline multimodal computed tomography imaging was reanalyzed to confirm that all the inclusion criteria were met. The radiological outcome was assessed by 2 independent neuroradiologists, and a third independent rater was consulted in cases of disagreement.

Statistical Analysis

Sample size calculation was primarily based on the EXTEND-IA TNK trial (Tenecteplase Versus Alteplase Before Endovascular Therapy for Ischemic Stroke) results,¹⁹ where 22% of patients receiving intravenous tenecteplase and 10% of those treated with intravenous alteplase reached major reperfusion. In addition, in both groups, 1% of patients developed sICH. Therefore, we hypothesized that 21% of patients in the tenecteplase group and 7% in the best medical treatment group would attain the primary outcome of major reperfusion without sICH. This trial design aimed for an 85% power to detect between-group differences with a 2-sided type I error of 0.05, accounting for a 5% dropout rate, resulting in a total of 224 patients (112 patients in each group).

The comprehensive statistical analysis plan is specified in the Supplemental Material and has been ratified and completed before the database lock. Analyses were executed using SAS version 9.4, with a 2-sided significance level set at 0.05. The primary analysis was conducted based on the intention-to-treat population, and the missing data remained missing. The comparison of the primary outcome proportions between the tenecteplase group and the best medical treatment group used modified Poisson regression with a log-link function and robust error variance. This was adjusted for prespecified covariates, including the interval from the last known well to randomization

(4.5–9 versus 9–24 hours), the site of vessel occlusion (internal carotid artery versus other vessels), and whether EVT was preplanned (yes versus no), with the presentation of adjusted relative risks alongside 95% CIs.

For the analysis of secondary radiological efficacy outcomes, the difference in recanalization rates was analyzed using modified Poisson regression with a log-link function and robust error estimation. This analysis was adjusted for prespecified factors: the interval from the last known well to randomization (4.5–9 versus 9–24 hours), the site of vessel occlusion (internal carotid artery versus other vessels), and whether EVT was preplanned (yes versus no). Differences in infarct growth volume between groups were evaluated using a median regression model with the adjustment of the interval from the last known well to randomization (4.5–9 versus 9–24 hours), baseline core volume, and preplanned EVT (yes versus no).

In assessing secondary clinical efficacy and safety outcomes, variables such as age, the interval from the last known well to randomization, baseline NIHSS, and preplanned EVT were entered as covariates. The differences in the proportions of patients achieving mRS scores of 0 to 1, 0 to 2, and 5 to 6 at 90 days, as well as major neurological improvement at 24 to 48 hours, were evaluated using modified Poisson regression with a log-link function and robust error estimation. For the ordinal analysis of the mRS distribution at 90 days, a Brant test confirmed the appropriateness of the proportional-odds assumption. Subsequently, an ordinal regression model was applied to compare the outcomes across the entire range of mRS, reporting a common odds ratio as an effect estimate for improvement for at least 1 point in the score. The analysis of change in NIHSS at 24 to 48 hours was conducted using a median regression model. Secondary radiological safety outcomes analysis used Poisson regression with a log-link function and robust error estimation, adjusted solely for preplanned EVT due to the low occurrence of rates of these outcomes, which limited the feasibility of multiple multivariable adjustments.

Subgroup analyses were conducted to investigate the impacts of various factors on the primary and preselected secondary outcomes. These factors included the site of occlusion (internal carotid artery versus others), the interval from the last known well to randomization (4.5–9 hours versus 9–24 hours), the type of stroke onset (witnessed versus unwitnessed stroke), and preplanned EVT (yes versus no). Specific outcomes of interest for these analyses encompassed recanalization rates observed in initial catheter angiography or 4- to 6-hour CTA reconstructed from CTP, infarct growth measured at 3 to 5 days, mRS score 0 to 1, mRS score 0 to 2, and mRS distribution at 90 days.

Sensitivity analysis was conducted for the primary outcome under a range of assumptions about missing data. The missing data on the primary outcome in the primary analysis were under the assumption of being missing at random. The results of plausible departures from missing at random were explored in the intention-to-treat population with a pattern-mixture model.

RESULTS

A total of 6205 patients were screened from October 21, 2021 to June 13, 2023, with 224 patients finally enrolled and randomized in this trial (Figure 1). The

3-month follow-up was completed by the end of October 2023. Of these, 111 patients were assigned to receive intravenous tenecteplase at a dose of 0.25 mg/kg, whereas 113 patients were allocated to the best medical treatment. The tenecteplase group had a median (SD) age of 64.2 (10.4) years, and the baseline median (interquartile range [IQR]) NIHSS score was 9.0 (5.0–14.0). A proportion of 53.2% (59) of patients were transferred to the catheter room for preplanned EVT, whose primary outcome was assessed through catheter angiography. Among them, EVT was averted in 1 patient due to early reperfusion (modified Treatment in Cerebral Ischemia $\geq 2b$) at the initial angiographic run. In the best medical treatment group, the median (SD) age was 63.6 (11.0) years, and the baseline median (IQR) NIHSS was 9.0 (6.0–16.0). In the best medical treatment group, 23.1% (26 patients) received intravenous alteplase, and 1.8% (2 patients) were treated with urokinase, including 1 case of intra-arterial thrombolysis. A proportion of 56.6% (64) of patients proceeded to the catheter room for preplanned EVT (Table 1). The reasons why participants did not transfer to the catheter room are listed in Table S1. Of the 123 patients transferred for preplanned EVT, 44 received adjuvant tirofiban, and 12 received heparin. The primary outcome could not be evaluated for 2 patients in the best medical treatment group due to the absence of follow-up perfusion imaging, allowing for primary outcome analysis in 111 patients per group.

The primary outcome, major reperfusion without the occurrence of sICH, was achieved in 37 (33.3%) patients in the tenecteplase group and 12 (10.8%) patients in the best medical treatment group (adjusted relative risk, 3.0 [95% CI, 1.6–5.7]; $P=0.001$). Sensitivity analysis concerning the missing data drew similar results (Figure S1).

In the analysis of efficacy outcomes, recanalization was achieved in 39 (35.8%) patients in the tenecteplase group and 16 (14.3%) patients in the best medical treatment group (adjusted relative risk, 2.5 [95% CI, 1.4–4.4]; $P=0.002$) at the initial catheter angiography or 4- to 6-hour CTA reconstructed from CTP. However, no statistically significant difference was found between treatment groups concerning other radiological and clinical efficacy outcomes, including infarct growth at 3 to 5 days, change in NIHSS at 24 to 48 hours, proportions of major neurological improvement at 24 to 48 hours, as well as proportions of mRS score 0 to 1, mRS score 0 to 2 and mRS distribution (Table 2; Figure S2).

In the analysis of safety outcomes, parenchymal hematoma type 2 at 24 to 48 hours occurred in 10 (9.0%) patients in the tenecteplase group and 4 (3.5%) patients in the best medical treatment group (adjusted relative risk, 2.7 [95% CI, 0.8–8.5]; $P=0.1$). No statistically significant difference was found between treatment groups concerning proportions of sICH, any intracranial hemorrhage at 24 to 48 hours, systemic bleeding at discharge, and mRS score 5 to 6 at 90 days (Table 2).

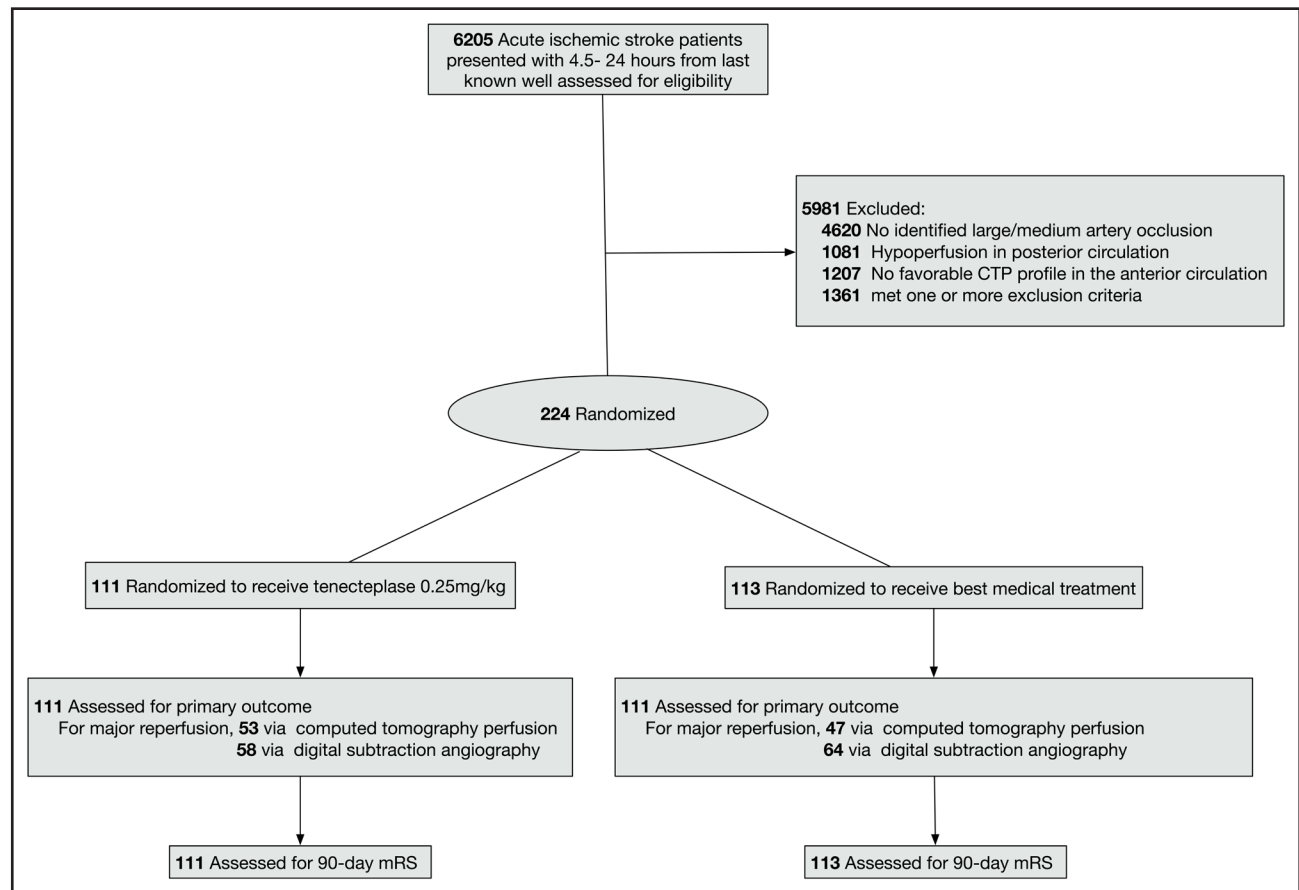


Figure 1. Trial profile.

CTP indicates computed tomography perfusion; and mRS modified Rankin Scale.

In the preplanned subgroup analysis, significant interactions were observed in subgroups of preplanned EVT regarding the primary outcome and recanalization. Tenecteplase significantly favors the achievement of the primary outcome and recanalization for patients undergoing preplanned EVT, compared with patients without EVT. In addition, there was a notable trend toward significance in the interactions between tenecteplase treatment and preplanned EVT in terms of infarct growth and mRS distribution. Interestingly, although tenecteplase positively influenced the primary outcome, it showed a tendency to increase the risk of larger infarct volumes and a shift in mRS distribution towards worse clinical outcomes among patients with preplanned EVT (Figure 2).

DISCUSSION

The CHABLIS-T II trial demonstrated that among patients with acute stroke with large or medium vessel occlusion(s) and favorable penumbral profile presented within 4.5 to 24 hours from the last known well, intravenous thrombolysis with tenecteplase at the dose of 0.25 mg/kg resulted in a higher incidence of reperfusion, compared with the best medical treatment without tenecteplase thrombolysis but that included planned EVT

and 23% of patients treated with intravenous alteplase. No difference was shown concerning clinical and functional improvement after treatment.

The higher reperfusion efficacy has already been acknowledged in patients with acute stroke with large vessel occlusion(s) presented within the 4.5-hour time window in randomized clinical trials and real-world practice.^{19–23} For patients in the extended time window, the performance of intravenous tenecteplase has been explored by our previous CHABLIS-T trial, the TIMELESS trial, and the TRACE-III trial.^{11–13} However, TIMELESS did not observe a higher incidence of reperfusion in the tenecteplase group after EVT, which might be explained by its short dwell time from thrombolysis to groin puncture (15 minutes). The proportion of primary outcome (major reperfusion without the occurrence of sICH) in the tenecteplase group of the CHABLIS-T II trial (33.3%) was very close to that of the 0.25 mg/kg tenecteplase group in our previous CHABLIS-T trial (32.6%), implicating a comparatively stable patient population during enrollment. Also same as the results of the CHABLIS-T trial, the proportion of reperfusion/recanalization was comparatively higher in the CHABLIS-T II trial than that in the EXTEND-IA TNK trials (33% versus 20%).^{19,24} Possible explanations

Table 1. Baseline Characteristics of the Patients Randomized

	Tenecteplase (n=111)	Best medical treatment (n=113)
Age, y, mean (SD)	64.2 (10.4)	63.6 (11.0)
Male sex, n (%)	80 (72.1%)	80 (70.8%)
NIHSS score at randomization, median (IQR)	9 (5–14)	9 (6–16)
Stroke cause, n (%)		
Cardioembolism	11 (9.9%)	20 (17.7%)
Large artery atherosclerosis	88 (79.3%)	83 (73.5%)
Undetermined cause	12 (10.8%)	10 (8.9%)
Medical history, n (%)		
Atrial fibrillation	9 (8.1%)	20 (17.7%)
Hypertension	71 (64.0%)	79 (69.9%)
Diabetes	32 (28.8%)	26 (23.0%)
Ischemic stroke or transient ischemic attack	15 (13.5%)	19 (16.8%)
TLKW to randomization, h, n (%)		
4.5–9	34 (30.6%)	37 (32.7%)
9–24	77 (69.4%)	76 (67.3%)
Witnessed stroke, n (%)	43 (38.7%)	43 (38.1%)
Transferred for preplanned endovascular treatment, n (%)	59 (53.2%)	64 (56.6%)
TLKW to hospital arrival, min, median (IQR)	595.0 (398.0–818.0)	525.0 (368.5–778.5)
TLKW to initiation of intravenous therapy, min, median (IQR)*	740.0 (524.0–954.0)	660.0 (470.0–817.5)
Time from hospital arrival to initiation of intravenous therapy, min, median (IQR)*	114.0 (79.0–160.0)	96.0 (78.5–113.0)
Time from hospital arrival to initial angiographic assessment, min, median (IQR)†	170.0 (122.0–232.0)	177.5 (119.3–230.0)
Time from initiation of intravenous thrombolysis to initial angiographic assessment, min, median (IQR)‡	40.0 (24.0–78.0)	60.5 (44.0–159.5)
Site of artery occlusion, n (%)		
Extracranial segment of internal carotid artery	16 (14.4%)	21 (18.6%)
Intracranial segment of internal carotid artery	9 (9.9%)	7 (6.2%)
First segment of middle cerebral artery	53 (47.8%)	55 (48.7%)
Second segment of middle cerebral artery	21 (18.9%)	24 (21.2%)
Anterior cerebral artery	10 (9.0%)	6 (5.3%)
Ischemic core volume at baseline, mL, median (IQR)	6.0 (2.0–25.0)	9.0 (3.0–22.0)
Hypoperfusion volume at baseline, mL, median (IQR)	87.0 (54.0–136.0)	90.0 (60.5–114.0)

IQR indicates interquartile range; NIHSS, National Institutes of Health Stroke Scale; and TLKW, time from the last known well.

*For patients in the best medical treatment group, n=29, including 2 patients crossed over to tenecteplase treatment.

†For patients in the tenecteplase group, n=59; for patients in the best medical treatment group, n=64, including 1 patient crossed over to tenecteplase treatment.

‡For patients in the tenecteplase group, n=59; for patients in the best medical treatment group, n=6, including 1 patient crossed over to tenecteplase treatment.

include a longer time from treatment to reperfusion assessment in CHABLIS-T trials, especially for patients without preplanned EVT assessed through 4-to-6-hour perfusion imaging, as well as the difference between imaging modalities applied for reperfusion assessment (digital subtracted angiography versus CTP), which has been thoroughly discussed in the previously published CHABLIS-T trial.¹³ Notably, the incidence of hemorrhagic transformation in both groups of CHABLIS-T II was numerically higher compared with that of the TIMELESS and TRACE-III trials.^{11,12} The difference between the TIMELESS and the CHABLIS-T II trials might be explained by our higher proportion of large artery atherosclerotic (LAA) stroke because early antiplatelet therapy would be needed if angioplasty

or stent implantation were performed. As for the discrepancy with TRACE-III, which was also conducted in the Chinese population who were dominated by LAA-related stroke, optional EVT might contributed to the higher bleeding rate in our trial, which needs further investigation.

No significant difference was shown between treatment groups in secondary outcomes, including NIHSS at 24 to 48 hours, infarct growth at 3 to 5 days, or mRS score at 90 days. Therefore, it seems that the improved reperfusion did not transform into radiological and clinical benefits in the tenecteplase group. Although CHABLIS-T II was not designed to demonstrate clinical efficacy and the sample size was underpowered, one of the major possible reasons for the neutral secondary outcome results may be

Table 2. Outcomes

	Randomized treatment group		Effect size	95% CI	P value
	Tenecteplase (n=111)	Best medical treatment (n=113)			
Primary outcome, n (%)	37/111 (33.3%)	12/111 (10.8%)	aRR: 3.0	1.6 to 5.7	0.001
Secondary outcome					
Efficacy					
Recanalization, n (%)	39/109 (35.8%)	16/112 (14.3%)	aRR: 2.5	1.4 to 4.4	0.002
Infarct growth at 3–5 d, mL, median (IQR)*	18.9 (4.2 to 59.6)	16.5 (2.6 to 42.3)	Adjusted difference: 5.0	–2.8 to 12.9	0.2
Major neurological improvement at 24–48 h, n (%)	24/110 (21.8%)	26/112 (23.1%)	aRR: 0.9	0.5 to 1.6	0.8
Change in NIHSS score at 24–48 h compared with baseline, median (IQR)†	–1 (–4 to 0)	–1 (–5 to 0)	Adjusted difference: 0.2	–1.0 to 1.4	0.7
mRS score 0–1 at 90 d, n (%)	44/111 (39.6%)	41/113 (36.3%)	aRR: 1.1	0.7 to 1.6	0.8
mRS score 0–2 at 90 d, n (%)	56/111 (50.5%)	66/113 (58.4%)	aRR: 0.8	0.6 to 1.2	0.4
mRS score at 90 d, n (%)			Adjusted cOR: 0.8	0.5 to 1.3	0.3
0	26 (23.4%)	23 (20.4%)			
1	18 (16.2%)	18 (15.9%)			
2	12 (10.8%)	25 (22.1%)			
3	17 (15.2%)	14 (12.4%)			
4	16 (14.4%)	13 (11.5%)			
5	10 (9.0%)	8 (7.1%)			
6	12 (10.6%)	12 (10.6%)			
Safety					
Symptomatic intracranial hemorrhage, n (%)	6/111 (5.4%)	5/113 (4.4%)	aRR: 1.3	0.4 to 4.2	0.7
Any intracranial hemorrhage, n (%)	27/111 (24.3%)	21/113 (18.6%)	aRR: 1.4	0.8 to 2.4	0.3
Parenchymal hemorrhage type 2, n (%)	10/111 (9.0%)	4/113 (3.5%)	aRR: 2.7	0.8 to 8.5	0.1
mRS score 5–6 at 90 d, n (%)	22 (19.8%)	20 (17.7%)	aRR: 1.1	0.6 to 2.1	0.7
Systemic hemorrhage, n (%)	5/111 (4.5%)	7/113 (6.2%)	aRR: 0.8	0.2 to 2.4	0.6
Severe or life-threatening	0 (0.0%)	0 (0.0%)			
Moderate	0 (0.0%)	0 (0.0%)			
Mild	5 (4.5%)	7 (6.2%)			

aRR indicates adjusted relative risk; cOR, common odds ratio; IQR, interquartile range; mRS, modified Rankin Scale; and NIHSS, National Institutes of Health Stroke Scale.
*n=102 for the tenecteplase group and n=109 for the best medical treatment group.
†n=110 for the tenecteplase group and n=112 for the best medical treatment group.

the characteristics of our patient population, which consists of around 76% of patients with LAA-related stroke, much higher than the EXTEND-IA TNK and TIMELESS trials.^{12,19,24} Only 56.6% of patients in the best medical treatment group underwent preplanned EVT but the proportion of functional independency (3-month mRS score, 0–2) was as high as 58.4%, higher than the tenecteplase and placebo group of TIMELESS trial, where around 77% patients underwent EVT. Despite that, the baseline NIHSS in CHABLIS-T II (median, 9) was lower than that in TIMELESS (median, 12),¹² the discrepancy between functional outcome might also be attributed to the different hemodynamic patterns of patients recruited. The penumbra mismatch criteria of this trial enabled the selection of the slow progressors with very robust collateral supply, which were mostly patients with LAA under the setting of the East Asian population.²⁵ Therefore, although more

patients with proximal artery occlusion included (internal carotid artery CHABLIS 24% versus TIMELESS 8%),¹² a naturally benign collateral flow pattern in atherosclerotic patients compared with cardioembolic patients partly contributed to the lower baseline NIHSS score in CHABLIS-T II.²⁶ In addition, although with a much lower rate of EVT, the infarct growth in CHABLIS-T II was only slightly larger than the infarct growth volume in the TIMELESS trial (CHABLIS-T II: tenecteplase 18.9 mL, control 16.5 mL; TIMELESS: tenecteplase 12 mL, placebo 16 mL).¹² Therefore, it is appropriate to deduce that there were part of patients with acute stroke, especially the ones with LAA-related stroke, with very robust and stable hemodynamic support whose penumbra barely grew into infarction even without reperfusion, namely the ultra-slow progressors, which needs further investigation. TRACE-III, however, managed to demonstrate the superiority of

tecteplase over the best medical treatment in Chinese patients with a sample size double of CHABLIS-T II. Notably, the proportion of patients achieving a 3-month mRS score of 0 to 1 or 0 to 2 in the best medical treatment group was significantly lower in TRACE-III compared with CHABLIS-T II.¹¹ This disparity might be explained by the use of different perfusion imaging postprocessing software (MISter and iStroke) and the allowance of endovascular therapy in CHABLIS-T II.

Futile reperfusion, an intensely studied phenomenon, was also unneglectable when interpreting the results of CHABLIS-T II. First, the major reperfusion in this trial was defined as the decrease in volume of the hypoperfusion lesion (DT >3 seconds). However, the restoration of

microcirculation blood flow was generally demonstrated as a decrease in cerebral blood flow.²⁷ Therefore, the decrease of DT >3 seconds volume might not guarantee a recovery from microcirculation ischemia. Second, various factors, such as stroke severity and hyperglycemia, were associated with increased odds of futile reperfusion,²⁸ which also needs further validation in our study.

Preplanned subgroup analysis revealed that intravenous tecteplase significantly favors patients with planned EVT in primary outcome and recanalization, which was rational because more patients with proximal large artery occlusion were prone to EVT. However, patients who had undergone preplanned EVT showed a trend towards a larger infarct growth and a worse

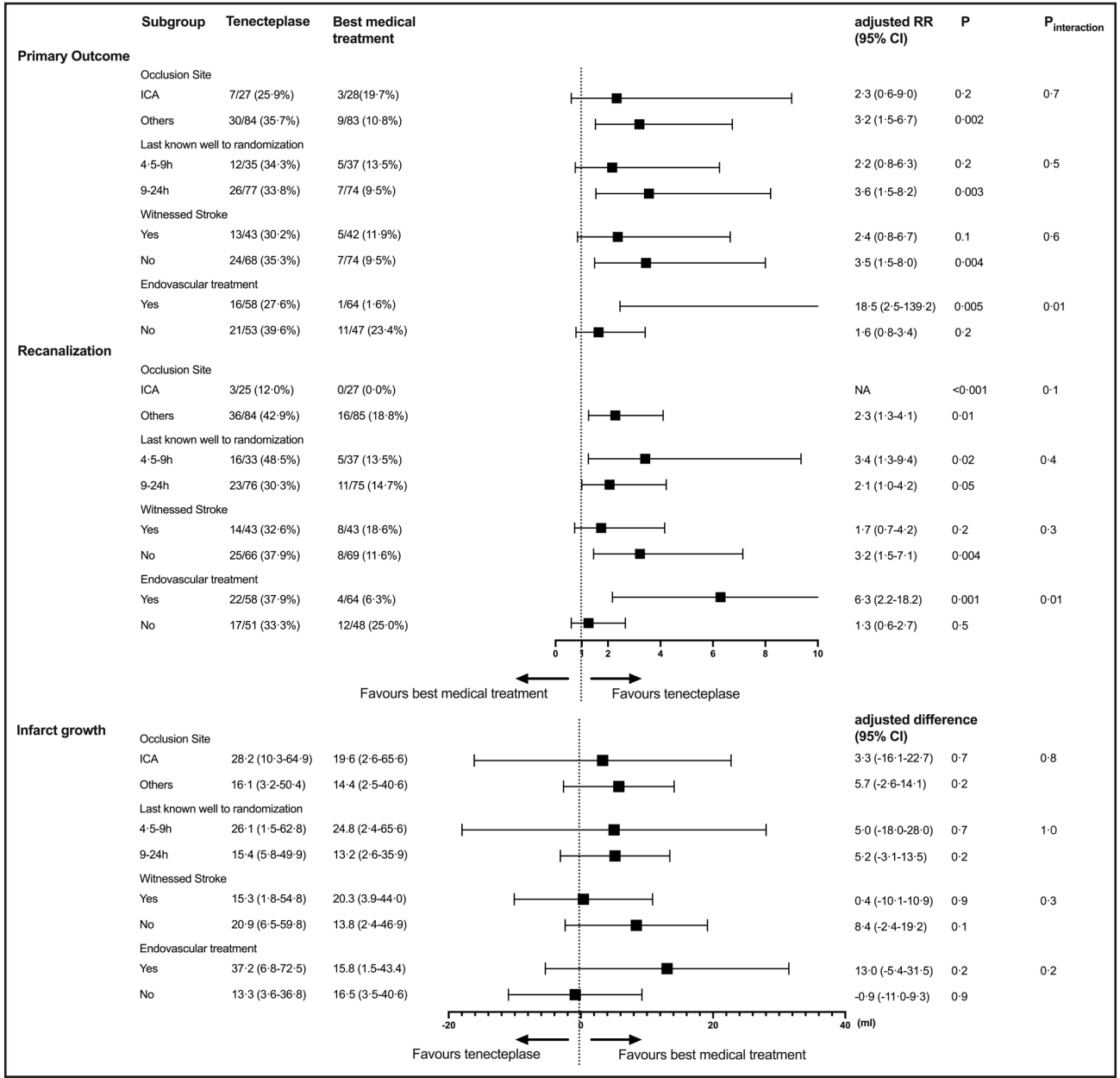


Figure 2. Subgroup analysis. For proportion of each subgroup, please refer to Table S2. ICA indicates internal carotid artery; and RR, relative risk.

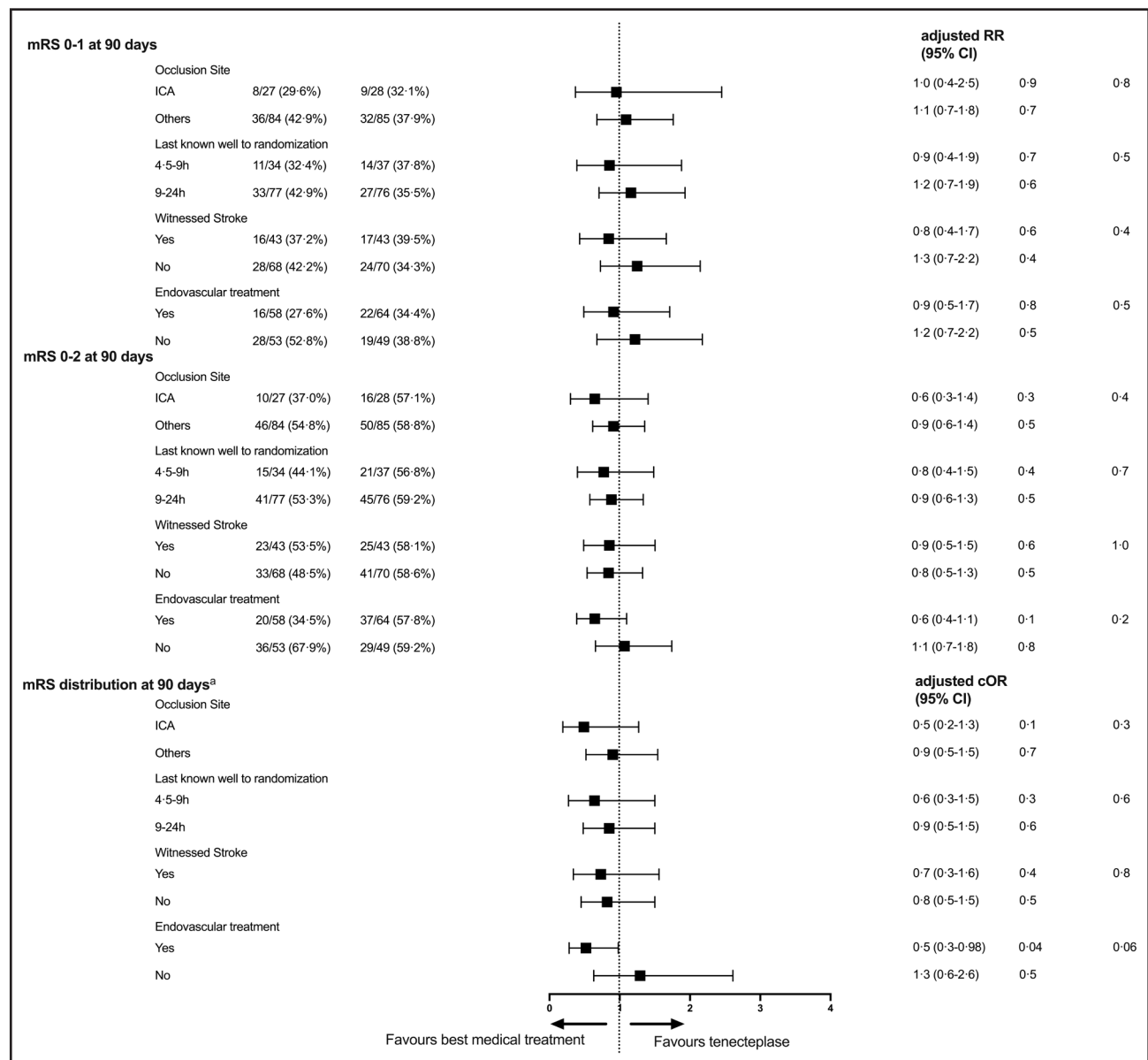


Figure 2 Continued.

functional outcome at 90 days (higher mRS score) when treated with intravenous tenecteplase, contrary to its performance on reperfusion and recanalization. The proportion of patients achieving functional independence with EVT treatment in the tenecteplase group was much lower than that in the TIMELESS trial (CHABLIS-T II 34.5% versus TIMELESS 46.0%).¹² One explanation to decipher these confusing results was that bridging therapy might increase the risk of hemorrhagic transformation in a patient population dominated by LAA because early antiplatelet therapy would be occasionally needed. In addition, occlusion due to intracranial or extracranial atherosclerosis is refractory to current stent retrievers or aspiration systems, therefore, a longer time-to-reperfusion might be needed to achieve recanalization, which was also associated with increased bleeding risk.²⁹

This possible higher bleeding risk partly contributed to the larger volume of infarct growth, since the calculation of infarct volume at 3 to 5 days harbored the volume of hematoma. Because this study was also underpowered to draw any conclusion from the subgroup analysis, the role of EVT and why tenecteplase treatment played a probably negative role in the functional outcomes in patients who had undergone EVT need further investigation.

CHABLIS-T II has the following limitations. First, the sample size was limited, so a composite safety and efficacy primary end point was set instead of long-term functional outcomes (mRS at 90 days). In addition, we expected the effect on the functional outcome of EVT to obscure any potential difference between tenecteplase and the best medical treatment including alteplase, which was shown in TIMELESS.¹² Second, the patients

enrolled were highly selected, only accounting for 3.6% of the candidates screened. Third, the criteria of hypoperfusion lesion in CHABLIS-T II was defined using DT >3 seconds rather than the widely applied time to maximum >6 seconds. Though previous studies have proven that the threshold of DT >3 seconds is as accurate as time to maximum >6 seconds when defining acute hypoperfused lesion,^{15,16} a post hoc analysis using time to maximum >6 seconds may be helpful to the generalizability of this study. Furthermore, this trial adopted a mismatch volume threshold of 10 mL and a ratio of 1.2, which allowed for the inclusion of a larger patient population compared with the more stringent criteria (mismatch volume of 15 mL and ratio of 1.8). As a result, the findings of the trial were more broadly generalizable. Fourth, reperfusion status was assessed through 2 imaging modalities, CTP or digital subtracted angiography, for patients with or without preplanned EVT. However, our previous study has proven that both imaging modalities could reliably assess the reperfusion status of patients with large vessel occlusion(s).³⁰ Fifth, only 54.9% of patients underwent preplanned EVT treatment. Major reasons for this low EVT rate included the inability to obtain informed consent and bleeding concerns from the neurointerventionalists. Sixth, contradiction remained to be explained concerning the analysis of outcomes, especially for the result that no difference in infarct growth was demonstrated between treatment groups, although tenecteplase induced a higher possibility of reperfusion. Seventh, time delay existed from thrombolysis administration to EVT, which might significantly impact the functional outcome of patients. Therefore, an improvement in the treatment workflow is needed.

In conclusion, administered at a dose of 0.25 mg/kg intravenously, tenecteplase increased reperfusion without sICH in patients with ischemic stroke selected by imaging in late-time window treatment but did not change clinical outcomes at 90 days. To translate sufficient and safe reperfusion into an improvement in functional outcome, other medical scenarios may be considered, such as using tenecteplase for thrombolysis at primary stroke centers followed by transporting patients to comprehensive stroke centers for EVT. Together with TRACE-III,¹¹ TIMELESS,¹² and the ETERNAL-LVO trial (Extending the Time Window for Tenecteplase by Effective Reperfusion in Patients With Large Vessel Occlusion; <https://www.clinicaltrials.gov>; Unique identifier: NCT04454788), CHABLIS-T II provides new evidence of intravenous tenecteplase treatment in patients with acute stroke in an extended time window.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S6
Figures S1–S2
Enrolling Sites and Collaborators
Protocol v1.0, v2.0, v3.0
Summary of Protocol Amendments
Statistical Analysis Plan
DSMB Charter
1st and 2nd DSMB Meeting Report

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